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To cite this article: Jinglu Li, Rebecca Currano, Dave B. Miller & David Sirkin (24 Apr 2025): A Framework for Proactive Interaction in Automated Vehicles, International Journal of Human-Computer Interaction, DOI: [10.1080/10447318.2025.2487719](https://doi.org/10.1080/10447318.2025.2487719)

To link to this article: <https://doi.org/10.1080/10447318.2025.2487719>



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Published online: 24 Apr 2025.



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A Framework for Proactive Interaction in Automated Vehicles

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ABSTRACT

As AI advances, and can draw upon previously disparate information sources, automated vehicles are gaining the ability to proactively collaborate with humans to accomplish everyday tasks. This collaboration will be based on increased sensing and information and will feature greater proactive behavior by the vehicle. However, there is little guidance on how vehicles should behave proactively, or even a shared understanding of what that means. Using prior work, we develop a framework of proactive interaction comprising three dimensions: initiation of action, alignment with users' goals, and communication of context awareness. We apply the framework in a video-based online study ($N = 351$) of a vehicle navigation system that exhibits proactive behavior. The system advises or acts, supports or counters users' goals in two scenarios (to reach a work meeting quickly or carry groceries without walking too far), and provides a more or less detailed rationale. We find that when goals are misaligned, communicating contextual awareness decreases user experience and acceptance. As a result, if alignment of (immediate) goals is not possible—such as when drivers want to reduce walking distance but the closest parking lot is full—systems should exercise care in communicating their situation awareness, as this can reinforce users' negative perceptions. Dispositional traits such as propensity to trust and locus of control do predict users' trust, distrust, and experience in our scenarios. We also find that participants' primary rationales for accepting the system's decision are largely similar (saving time, trusting, and making sense) but rationales overall, especially for rejecting the system's decision, are more varied than expected and often based on experience. Therefore, systems that look to personalize decision-making—such as whether to proactively execute versus merely suggest an action—could take into account users' dispositional traits and other preferences, such as those exemplified by these rationales.

KEYWORDS

Proactive interaction; automated vehicles; goals; communication; agency; technology acceptance; user experience; trust

1. Introduction

In the context of automobiles, increased access to data from sensors, and improved analytical capabilities of tools such as machine learning, are enabling automated systems to take more independent action. They might collaborate with humans to support their goals, or correct their bad decisions, or even act in ways that appear to counter their decisions, but which are based on information their human counterparts are unaware of and support their best interests. However, making informed design decisions about how these systems should behave—in the aggregate as well as specific cases—requires our understanding what proactive behavior really means. The goal of this paper is to propose a framework for studying *proactive interaction* which can be used to better organize discussion about proactivity in the context of automated vehicle (AV) interaction design, and to guide future research in this area.

With this inspiration, we formed four research questions to guide our exploration of how to define, scope, apply, and discuss proactive interaction. The first two guide us in

developing a framework for proactivity within the context of AV interaction design, while the next two address applying this framework in a situated driving scenario.

We conducted a literature review to survey prior thinking on people's perception of proactivity in HCI, HRI, and human vehicle interaction, as well as some foundational work from organizational behavior that grounded the research in these other areas. From this, using an inductive coding approach, we derived our three-dimensional model, which can serve as a first iteration of a framework for understanding proactive interaction. This model supports the development of tractable study designs that can yield sufficient statistical power with a reasonable number of respondents, and can thus be useful for empirical research. We evaluated this framework with an online survey study with 351 respondents, which supports these as being relevant factors in user perception of proactive automation. We cannot claim our model to be exhaustive, but our study findings support these dimensions being relevant constructs.

1.1. Research questions to frame reactive interaction

To advance knowledge within the human-AV research community, we begin by establishing a shared basic understanding of proactive interaction. This understanding should include definitions, dimensions that characterize (and thus allow researchers to manipulate) interactions, and how proactivity contributes to system performance and user experience. We thus start with the following research questions:

- RQ1: What does proactivity mean in the context of human-AV interaction design?
- RQ2: What dimensions are important to consider in characterizing proactive interaction?

In Section 2, we review prior work in automotive and other interaction design contexts that explore the meaning and application of proactivity (RQ1), and identify its basic and shared concepts (RQ2). Based on these concepts, in Section 3 we present a framework to define and understand proactivity in human-AV interaction design.

1.2. Research questions to apply reactive interaction framework

To apply the resulting framework, we conducted a study in the context of vehicle navigation. The study allows us to test whether and how the dimensions discovered (RQ2) provide meaningful insights that can guide design efforts. In our investigation of prior research on proactivity across disciplines, including human-human work (Allen et al., 2005; Parker et al., 2006), human-technology interaction (Kraus et al., 2021), and interaction with novel technologies such as robots (Chien et al., 2024; Zafari & Koeszegi, 2021) and automated vehicles (Choi & Ji, 2015) in particular, traits such as an inclination to trust or to perceive events as within or outside of one's control Lumpkin (1985) appear often. These two concerns naturally fit together, since proactive systems by their nature assume control otherwise held by humans, and humans in turn require some degree of trust in these systems to release control to them. And yet, these topics are not often explored together.

To better understand this interplay, we specifically chose to examine these two dispositional traits of propensity to trust (Frazier et al., 2013) and locus of control (Lumpkin, 1985). The propensity to trust reflects an individual's general willingness to trust others, including technology, which is particularly relevant in the context of automated vehicles, where users must entrust their personal safety to the vehicle's actions. The locus of control refers to the extent to which individuals believe they have control over the outcomes in their lives, or conversely, that external forces determine these outcomes. This trait is relevant as it can influence how users perceive and interact with automated vehicles, potentially affecting their experience with and acceptance of the technology. As both traits influence the way individuals approach new experiences and systems, they are crucial for understanding users' perspectives on engaging

with proactive systems. We ask the following research questions to guide our study applying the framework:

- RQ3: How might varying the dimensions that characterize proactivity influence users' responses to the system?
- RQ4: How might users' dispositional differences influence, or interact with, their responses to the system?

In Sections 4–6, we present an online study that applies multiple levels of these dimensions (RQ3) and integrates measures of the dispositional traits (RQ4) in an everyday driving context of navigation. Taken together, Sections 2–3 address RQ1 and RQ2 and Sections 4–6 address RQ3 and RQ4. Sections 7–9 then discuss and contextualize findings.

1.3. Contribution

This paper's contribution is threefold: (1) we review prior research in several disciplines that implies, defines, or applies the concept of proactive interaction, (2) we establish a framework for understanding and designing proactive interactions in the context of vehicle automation, and (3) apply the framework in an empirical study of proactive navigation in autonomous driving scenarios, as an example of the kinds of analysis and insights it supports.

2. Prior perspectives on proactive interaction design

Researchers in many disciplines have explored proactive interaction design through different lenses, with each drawing upon their area's corpus of prior work and terminology. While we focus on the context of human-AV interaction here, our literature review found deep and broad explorations in human-computer interaction (HCI), human-robot interaction (HRI), and organizational studies, and we include this relevant work to explore common themes that are found across areas.

2.1. Human-vehicle interaction

Prior research on proactive interaction in the context of vehicles reveals diverse perspectives on its definition, components, and measures. Several works explore the use and effects of proactive behavior without offering an explicit and distinctive definition, or establishing its primary constituent dimensions, while others (Kabtoul, Martinet, et al., 2020) note how little explicit attention has been paid to the term *proactive* in the literature. For example, in their study on proactive pedestrian-vehicle cooperation in lateral and frontal crossing scenarios, Kabtoul, Spalanzani, et al. (2020) describe the vehicle as acting “*reactively or proactively*” (p. 6959), yet offer no further description of either term. Árnason et al. (2014) developed a context aware multimodal proactive recommender system focused on presenting social, email, and news information. They aim to create a system with “*no need for interactivity*” (p. 99), and conclude by considering if the system could be improved by making it

“more aware of the user” (p. 110) by including personal driving profiles or observing drivers’ actions.

In his call for research into proactive computing, Tennenhouse (2000) contrasts proactive with interactive computing, which deliberately places humans “in the loop,” in that proactive systems will (1) directly sense and act upon the world, (2) exist in vast numbers and act at faster-than-human speeds, and (3) therefore necessitate moving humans “above the loop” (p. 44). Over several studies, Bader et al. (2010, 2011) develop and examine in-car proactive recommender systems which offer drivers points-of-interest such as fuel stations, based on a situation awareness model (Endsley, 1995) that accounts for relevant past and future situations. Their approach builds upon Tennenhouse’s characterization of proactive systems, where instead of answering user queries interactively, the system “works off tasks and is supervised by the user” (Bader et al., 2010, p. 2). Their complementary study of user acceptance of proactive recommender systems found that “situation explanations and their comprehensibility” and recommendations that are “useful for the subjects in their situations” influence perceived usefulness. They also found that “clear information delivery and trust” were crucial for the acceptance of in-vehicle recommendations (Bader et al., 2011, p. 47). Taken together, while these works do not explicitly define proactivity, they implicitly characterize it as being distinct from reactivity or interactivity and place the user in a supervisory role.

The works that do explicitly define proactivity overlap to varying degrees, yet lack a uniform set of features that compose it. For example, in writing about context aware recommender systems in vehicles, Vico (2013) describes a proactive system as one that “pushes recommendations to the user when the current situation seems appropriate, without the need for an explicit user request” (p. 2). His model accounts for user, temporal, geographic, and social contexts in making suggestions. Smirnov et al. (2015) build off earlier perspectives of proactive interaction with mobile devices in entertainment and travel contexts (Church & Smyth, 2008; Ricci, 2010), which they describe as offering recommendations “without user request, on the basis of the current situation and user profile analysis” (p. 4). They emphasize the benefit of predicting user behavior by “analyzing drivers’ needs, preferences and intentions” (p. 1), recognizing context, and understanding the situation. To that end, they propose a case study exemplifying their approach, based on a navigation-based parking scenario, which would analyze cost, distance, and availability of nearby parking spaces. The proposed analysis would also include driver preferences, schedule, and other relevant contextual information, such as traffic and weather. While the preceding works agree in that proactive systems should preclude explicit user requests, they differ in the contextual details that qualify their definitions.

2.2. Human-computer interaction

Similar to the literature in human-vehicle interaction, several works in human-computer interaction (HCI) frame proactivity

in terms of the system acting independently or without prior human prompting. For example, Nothdurft et al. (2015) explore key challenges for proactive dialogue systems and their influence on turn-taking behavior. They describe proactivity in technical systems as “an autonomous, anticipatory system-initiated behaviour, with the purpose to act in advance of a future situation, rather than only reacting to it.” Their framing also distinguishes interactive from proactive systems, which are “induced by implicit information and not by any kind of direct or explicit user interaction” (p. 74). Xiao et al. (2003) look at users’ reactions to anthropomorphic interface assistants comparing proactive agent, reactive agent, and paper manual conditions. They define proactivity as meaning “the assistant, in addition to answering queries, will make unsolicited suggestions” (p. 2) and note that action at the proper moment is an important feature for proactive agents. They also examine the role of users’ personality traits (McCrae & Costa, 1987) in their reactions, but find no significant correlations. And in an effort to map out and characterize the design space of adaptive interfaces, Isbell and Pierce (2005) introduce an interface-proactivity (IP) continuum, which frames proactivity primarily in terms of the responsibility for taking action. The continuum ranges from exclusively user driven (classical HCI systems) to exclusively system controlled (artificial intelligence systems). Associated metrics include the difficulty of use, cost of failure, intrusiveness, and the need for personal information about the user, all of which increase with autonomy.

Other literature extends the concept of proactivity to include qualifying attributes such as detecting or anticipating the context of use, and acting based on an understanding of users’ goals. Ju and Leifer (2008), for instance, develop a framework where proactive interactions are “initiated by the system” and reactive interactions are “initiated by the user” (p. 75), but they extend these notions through an exploration of foreground and background actions, attentional demand, settings, roles, and interactant initiative. Melguizo et al. (2007) study proactivity in the context of just-in-time text recommendations for professional writers through a system that aims to be unintrusive during various stages of the writing task. In their view, a proactive system should (1) present relevant information, (2) identify the author’s stage of writing and discern when the information is most needed and least disruptive, and (3) serve as an external long-term memory. Salovaara and Oulasvirta (2004) develop a user-centric typology of proactivity, focusing on several modes of resource management. They describe proactive systems as acting on the user’s behalf, and on their own initiative, qualifying that they are “intended to cause changes, rather than just to react to changes” (p. 58). The authors complement this definition with a list common attributes, including real-time operation, world model, hypothesized user goal state, sensitivity to future alternatives informed by users’ goals, and taking initiative toward those goals. They summarize the distinctive feature of proactivity as “mostly being deliberation of actions with respect to user’s goal state in the situation” (p. 60).

2.3. Human-robot interaction

Peng et al. (2019) note the lack of a clear concept of proactivity in HRI, finding the closest term to be autonomy as defined by ISO: “the ability to perform intended tasks based on current state and sensing without human intervention” (ISO 8373:2012(en), 2012). While some works do explicitly define proactivity, often using one dimension, others rely on an implicit understanding, such as framing it in contrast to reactivity. In a study of collaborative tasks with robotic drawers that vary expressivity and proactivity, Mok et al. (2015) define a proactive robot as one that “anticipated a user’s impending needs and initiated an action to complete a task” and contrast it with a reactive robot that “responded and executed actions according to the user’s gestures” (p. 700). Fink et al. (2014) study of a robotic toy box explores whether proactivity can motivate tidying behavior, and considers a proactive robot to be one that “moves toward a toy” while a reactive robot “responds to actions but more or less remains at one spot in the room” (p. 440).

Some researchers refine the concept of proactivity, expanding one dimension into multiple levels or including additional dimensions, factors, or measures. Drawing on research in organizational science, Peng et al. (2019) propose a framework comprising anticipation and autonomy dimensions. They define a service robot’s proactivity as “anticipatory action that robots initiate to impact themselves and/or others” with three elements of focus: “1) anticipation; 2) initiation of action; 3) target of impact” (p. 2), where anticipation is framed with respect to a human user’s needs. Meurisch et al. (2020) examine users’ expectations of and preferences for proactive AI systems. Drawing from Isbel and Pierce’s IP continuum, they categorize AI systems based on three classes (reactive, proactive, and autonomous), with the proactive class subdivided into two further levels (inform and recommend). Likewise inspired by Isbell and Pierce’s IP continuum and Peng et al.’s three levels of proactivity, Kraus et al. (2020) study the effects of varying proactivity and timing on a do-it-yourself assembly planning task. Measures include user experience, personality traits, and trust, and confounding variables include pre-disposed trust in automated systems, negative attitudes toward robots, and prior experience with similar systems and tasks.

2.4. Organizational behavior

The research in organizational science examines proactivity in the context of human behavior, and accordingly, it emphasizes work situations and includes deeper consideration of personality traits. For example, Bateman and Crant (1993) develop and assess a self-report scale of proactive tendencies using the Big Five personality factors (Costa & McCrae, 1989), dispositional measures of locus of control and needs for achievement and dominance, and criterion variables including extracurricular activities, personal achievements, and leadership. They consider the essential characteristic of proactive behavior to be a

person’s ability to “intentionally and directly change their current circumstances” (p. 104). In a review of proactive behavior in organizational contexts, Crant (2000) describes four constructs related to proactive behavior: proactive personality, personal initiative, role breadth self-efficacy, and taking charge. Each of these in turn represents an individual’s tendency to actively influence the environment and make change, take action that goes beyond job requirements to achieve goals, believe in one’s ability to perform a wider range of tasks beyond a defined role, and bring about constructive change in the workplace. He further defines proactive behavior as “taking initiative in improving current circumstances or creating new ones” and contrasts this with “passively adapting to present conditions” (p. 436). Parker and Collins (2010) review of literature found three common elements present in most definitions of proactive behavior: acting in anticipation, taking control, and self-initiation. And using similar wording to what we observed in other disciplines, Grant and Ashford (2008) develop a generalizable framework of employee proactivity in work settings and define proactive behavior as “anticipatory action that employees take to impact themselves and/or their environments” (p. 8). They contrast proactive against “more general motivated behavior and more reactive, passive behavior” (p. 8) in that it involves planning and acting in advance, with the intention to influence, change, and alter individuals, groups, and their environments. While these examples do not cover the entire breadth of organizational research, they do reflect its shared and distinctive concepts of proactive behavior, which we draw upon in the development of our three-dimensional framework.

3. A proactive interaction design framework

3.1. Grounding the framework

We approach our exploration of the dimensions of proactivity by analyzing prior work and application studies across multiple disciplines, which offer insight into proactive interaction from different perspectives. Even with these differences, our review of their definitions, dimensions, and features reveals commonly occurring themes both within and across disciplines. Using a reverse snowball strategy (Wohlin, 2016), we identified a list of key papers that discuss and define proactivity in interactive systems, as well as fundamental papers on proactivity in human behavior, which informed these.

After our initial review, we applied Thomas (2003) inductive analysis coding process, adapted from Creswell (2002). This involved first extracting concepts related to proactive interaction from descriptions in the literature (Table 1, column 2), then labeling these (column 3). The next part of the process iteratively reduced redundancy and removed any overlap until the 3–8 most important categories remained (as prescribed by Thomas) to form a model of proactive interaction. Table 2 shows the first set of categories (column 1), resulting from the first iteration, and the final three categories (column 2) that comprise our model,

Table 1. Descriptions and initial coding of proactive interaction related concepts.

Name	Proactive related description	Initial coding
Bateman and Crant (1993)	(Proactive Behavior) "a dispositional construct that identifies differences among people in the extent to which they take action to influence their environments"	Take action; Influence environment
Tennenhouse (2000)	(Proactive Computing) "us[es] sensors and actuators to both monitor and shape their physical surroundings," "routinely respond[s] to external stimuli at faster-than-human speeds," "[incorporates] modes of operation in which humans are above the loop"	Sense & act on world; Faster than humans; User supervision
Crant (2000)	(Proactive Behavior) defined as "taking initiative in improving current circumstances or creating new ones; it involves challenging the status quo rather than passively adapting to present conditions."	Take initiative; Change circumstances or conditions; Active
Xiao et al. (2003)	(Proactive Interface Assistants) "the assistant, in addition to answering queries, will make unsolicited suggestions" "Researchers must work to improve techniques for automated analysis of user goals and intentions, and then design interface agents that suggest actions only when confidence is extremely high that the advice will be worthwhile"	Make suggestions; Unsolicited; Contextual situation; User goals
Salovaara and Oulasvirta (2004)	(Proactive Systems) "proactive refers to two critical features of a system: 1) that the system is working on behalf of (or pro) the user, and 2) is taking initiative autonomously, without user's explicit command."	On behalf of user; Take initiative; Autonomous
Isbell and Pierce (2005)	(Adaptive User Interfaces) a continuum of increasing responsibility for taking action (e.g., unspecific alerts, suggestions, decisions) "need to be aware of and able to adapt to the particular needs of specific users"	Responsibility; Taking action; Aware of user needs
Melguizo et al. (2007)	(Proactive Recommendation Systems) (PRSs) "retrieve large quantities of documents, decide what available information is most likely relevant to the text to be written, and offer that information without user requests."	Retrieve information; Decide what is relevant; Inform without user requests
Ju and Leifer (2008)	(Interactive Systems) "interactions initiated by the system are proactive interactions"	System-initiated
Grant and Ashford (2008)	(Proactive Behavior) "anticipatory action that employees take to impact themselves and/or their environments"	Anticipatory; Action; Impact
Bader et al. (2010)	(Proactive In-Car Recommendations) "In contrast to interactive systems the user in proactive systems is not in the loop but above it"; "able to react before a situations occurs" "Context plays a fundamental role"	User supervision; Anticipate
Parker and Collins (2010)	(Proactive Behavior) "self-initiated, anticipatory action that aims to change and improve the situation or oneself"	Self-initiated; Anticipatory; Action; Change situation
Vico (2013)	(Proactive Recommender System)"Pushes recommendations to the user when the current situation seems appropriate, without the need for an explicit user request";	Pushes recommendations; Context-aware; Without user request
Fink et al. (2014)	(Proactive System) "tries to initiate an interaction (system driven) while in a reactive system, the user would take the initiative (user driven)."	System-initiated
Mok et al. (2015)	(Proactive Robot) "anticipate[s] a user's impending needs and initiate[s] an action to complete a task"	Anticipate; User needs; Initiate action; Complete task
Smirnov et al. (2015)	(Proactive Recommendation Systems) "offer recommendations ... without user request, on the basis of the current situation and user profile analysis"	Offer recommendations; Without user request; Current situation; User profile
Nothdurft et al. (2015)	(Proactivity in Technical Systems) "an autonomous, anticipatory system-initiated behaviour, with the purpose to act in advance of a future situation, rather than only reacting to it."	Autonomous; Anticipatory; System-initiated; Act in advance
Peng et al. (2019)	(Service Robots) "proactivity can be defined as the anticipatory action that robots initiate to impact themselves and/or others. The definition indicates three elements of robot's proactivity: 1) anticipation; 2) initiation of action; 3) target of impact."	Anticipatory; Initiate action; Impact
Kabtoul, Martinet, et al. (2020)	(Proactive Navigation) "the autonomous vehicle [is integrated] in the shared space as an influential agent, taking an active role in the scene and not merely reacting to it."	Shared space; Influential agent; Taking action
Meurisch et al. (2020)	(AI Systems, Proactive Support) "take actions without explicit user request. We further divide ... into ... two levels: (1) ... inform users what to pay attention to ... and (2) ... make personalized recommendations"	Take action; Without user request; Inform users; Make personalized recommendations
Kraus et al. (2020)	(Proactive Behaviour) "is about taking control, anticipating and preventing problematic situations instead of watching things happen"	Take control; Anticipate; Prevent problems
Tan et al. (2020)	(Proactive Behavior) "includes ... anticipation of human intention ... which includes information retrieval and context consciousness that predict the environment and behavior information and choose the interaction strategy independently to change the completed environment state."	Anticipation; Human intention; Context consciousness; Choose strategy independently; Change the environment
Zafari and Koeszegi (2021)	(Human Robot Collaboration) "In a collaborative context, a robot that demonstrates proactive behaviors seems to have more agency compared to situations in which the robot only responds to orders (i.e., reactive behavior)."	Collaborative; Agency
Chien et al. (2024)	(Human-Robot Interaction) "A proactive behavior can be defined as a robot initiating an anticipatory action ... [as opposed to] the robot reacting to a user request."	Initiation of action; Anticipatory

from the second iteration. The final category model includes: agency and taking initiative, anticipating user needs or goals, and understanding and communicating context.

3.1.1. Agency and taking initiative

The most basic and common feature of proactivity in the prior literature is a system's ability to initiate behavior that

Table 2. The categories list after inductive analysis.

First Set of Categories	Final Category Model
User supervision; Take initiative; Initiate action; System-initiated; Anticipatory; On behalf of users	Agency and taking initiative
User goals; Aware of user needs; User profile; Human intention	Anticipating user needs or goals
Influence environment; Sense & act on world; Change Circumstances/Conditions; Contextual situation; Context consciousness; Context-aware	Understanding and communicating context

enacts change in the user or the environment, a concept we refer to as *agency*.¹ Several earlier works distinguish proactive from reactive (Fink et al., 2014; Kabtoul, Martinet, et al., 2020; Meurisch et al., 2020) or interactive (Árnason et al., 2014) action, and place the user outside of the initiation of behavior (Bader et al., 2010, 2011; Tennenhouse, 2000). Others discuss the ability to push information to the user without prior request (Isbell & Pierce, 2005; Ju & Leifer, 2008; Smirnov et al., 2015; Vico, 2013; Xiao et al., 2003), or to act in anticipation of a future situation (Mok et al., 2015; Nothdurft et al., 2015; Peng et al., 2019). Where researchers describe degrees of proactivity, they consider enacting change (Bateman & Crant, 1993; Crant, 2000; Grant & Ashford, 2008; Peng et al., 2019; Salovaara & Oulasvirta, 2004) to be more proactive than making suggestions.

3.1.2. Anticipating user needs or goals

Concepts of proactivity frequently emphasize the importance of anticipating the user's needs or goals, to inform and direct the initiation of action. Some works look to analyze users' needs and understand the context as a means to choose among actions and their timing (Ju & Leifer, 2008; Mok et al., 2015; Peng et al., 2019; Salovaara & Oulasvirta, 2004), while others develop driver profiles to better understand users, which can lead to better personalized recommendations (Smirnov et al., 2015). Árnason et al. (2014) pointed out that because irrelevant information is distracting, intelligent systems should understand content and its relevance to the user, and present information in a personalized way, potentially applying personal profiles.

3.1.3. Understanding and communicating context

The implementation of proactive behavior, especially in human-vehicle interaction research, is often supported by (1) developing an awareness of the context of the system's use, and (2) selecting and communicating aspects of that information back to the user. In recognizing context, systems may observe users' actions (Árnason et al., 2014), account for cues in the physical setting (Ju & Leifer, 2008), or identify the current stage of an ongoing process (Melguizo et al., 2007). They base recommendations or actions on situation awareness (Bader et al., 2010) or world (Salovaara & Oulasvirta, 2004) models, user-temporal-geographic-societal context (Vico, 2013), or recognition of the current situation and broader environment (Smirnov et al., 2015). Communicating this basis then provides transparency, clarity, or comprehensibility to the user, and supports trust in the system (Bader et al., 2010, 2011), and improves user attitudes and performance (Koo et al., 2015). Ju and Leifer (2008) add that behavior can also signal intent, even without making explicit statements.

3.2. Defining the framework

Building on these themes, and to answer RQ1 and RQ2 in the context of human interaction with intelligent vehicle systems, we define proactivity as *behavior that takes initiative (such as offering advice or taking control) without prior prompting, based on an evaluation of the current situation, which includes anticipating the user's needs or goals, and understanding the environmental context*. Communicating the system's knowledge of user goals and/or environmental context can support the user's engagement with the system, and influence responses such as trust, acceptance, or how the user experiences the interaction. We operationalize this definition using three dimensions:

- *Initiation of action*: The level of independent behavior, expressed through advice, action, or control to effect change, initiated by the system.
- *Goal alignment*: The relationship between the system's behavior, including its decisions, communications, or actions, and the user's needs or goals.
- *Communication of context awareness*: The information the system has and communicates about the current or anticipated state of the user, environment, or situation.

We propose that these dimensions capture much of the breadth of prior research on proactivity, and that designers can vary the behavior of automated systems along these primary dimensions to create socially and situationally appropriate interactions.

In Sections 4–5, we describe how we applied this framework in an empirical study, involving an online video-based survey with 351 participants, that serves as a practical application of the framework. The study tests the framework in a situated context of vehicle navigation, examining how varying the dimensions of proactivity influence user responses. This application demonstrates how the framework can be used to generate insights and understanding in a real-world context, and therefore serves as a verification (Oreskes et al., 1994) of our model.

4. Exploring proactivity in a situated context

To explore if our framework supports the study of proactive behavior in automated vehicles settings, and to answer RQ3 and RQ4, we designed an experiment that casts participants in the role of drivers interacting with a vehicle's proactive navigation system. The study was approved under Stanford University's Institutional Review Board protocol #38139 titled *Evaluating Interaction Design through Online Audio and/or Video*.

We begin from the premise that proactive systems, by their nature, exhibit agency, as all of the works in [Section 2](#) implicitly or explicitly include independent initiation of action as a dimension, and many explore this dimension alone. While we agree on the primacy of initiation of action in characterizing proactivity, we also believe that (in the context of interaction design) goal alignment and communication of context color users' experiences and responses at least as much or more than agency does. The study therefore asks how varying these three dimensions might affect users, and looks to understand these changes through observable measures such as:

- To what degree users agree with the system's suggestions or actions, and whether they will accept or reject them
- To what extent users feel that the system is trustworthy
- How users will experience interacting with the system, based on its alternative behaviors

The scenario shares features with Smirnov et al.'s proposed system (Smirnov et al., 2015), including proactive navigation support to select a parking spot, sensitivity to the driver's schedule, and awareness of the immediate road context. But while Smirnov et al. focus on how a prototype system might function, we explore how participant drivers react to alternative system behaviors under varying conditions.

4.1. Hypotheses

As design researchers, we favor an exploratory approach when forming and investigating a new framework, so we developed three multipart hypotheses, based on expectations from the prior work and how the above dimensions—independently and in combination—might affect users' experiences and reactions.

- *H1: Direct effects of dimensions.* For each of the three dimensions of proactivity, independently increasing (a) alignment between the system's behavior and the user's goals, (b) communication of the system's awareness of the environment, or (c) initiation of action, will result in greater agreement with the system and acceptance of its behavior, better user experience, and higher levels of trust. (RQ3)
- *H2: Interaction effects of dimensions.* Variation of the dimensions together will amplify the effects observed. Specifically, a system whose behavior is misaligned with the user's goals will worsen agreement, acceptance, user experience, and trust scores when it also (a) communicates a greater awareness of the context, or (b) acts rather than suggests action. Additionally, (c) a system that acts rather than suggests action will improve scores on these measures when it also communicates greater context awareness. (RQ3)
- *H3: Effects of dispositional traits.* (a) *Direct effects:* users who tend to perceive events as outside of their control will be more inclined to agree with and accept the

system's actions, and those who have a greater propensity to trust technology will have a better experience and will be more inclined to trust the system. (b) *Interaction effects:* increased tendency to perceive events as outside of one's control, and inclination to trust technology, will amplify the effects of the system's behavior on the dependent measures. (RQ4)

The three hypotheses thus map to each of the three dimensions of proactivity, along with two dispositional traits, to observable measures of interaction quality. These are not meant to be an exhaustive list of all possible hypotheses that could arise from the research questions, rather they are intended to demonstrate the usefulness of the framework for exploring the practical effects of varying its constituent parts in the automated vehicles context.

5. Study design

The experiment presents a proactive navigation system that considers the available parking context, and drivers' goals regarding distance and time, across four scenarios, reminiscent of Smirnov's proposed case study (Smirnov et al., 2015). It has a between-participants, full-factorial design, comprising three independent variables (the three dimensions of proactivity from [Section 4](#)) with two levels each: alignment between the system's suggestion or action and the driver's goals (low vs. high), communication of contextual awareness (low vs. high), and the system's level of agency (suggest vs. act), resulting in eight possible combinations. Dependent variables include: how much participants agree with the system's suggestion or action, whether they would accept or reject its action, and evaluations of both user experience and trust. Given the potential for preexisting traits to influence responses, particularly for the latter two evaluations, we include covariate measures for locus of control and propensity to trust. Each of these variables is described in detail below.

5.1. Independent variables

5.1.1. Goal alignment

To explore the influence of goal alignment, we created two scenarios. In one, the participant is driving to the grocery store, and will have to carry several bags back to the car after shopping, and thus has the goal to minimize walking distance. In the other scenario, the participant is driving to work for an important meeting, but is running late, cueing time pressure, and thus has the goal to reach the destination as soon as possible. In both scenarios, while still a few blocks away from the destination, the navigation system warns that the nearest parking lot is full, informs that another lot is available one block further from the destination, and sets its goal to park there. Thus, in the grocery scenario, the system contradicts the driver's goal of minimizing distance; while in the work scenario, it aligns with the driver's goal of minimizing time.

Table 3. Spoken messages that the system communicated to drivers, for each level (low versus high) of the three study conditions.

Condition	Level	System's Spoken Message
Goal Alignment	Low	"You are going to the grocery store to shop for the week. You will need to carry several bags back to your car." (The driver's goal is to minimize walking distance. The system will contradict the driver's goal.)
	High	"You are going to the office for an important work meeting. Unfortunately, you are running late." (The driver's goal is to minimize time. The system will support the driver's goal.)
Context Awareness	Low	"The nearest parking lot is full. Another lot is available."
	High	"The nearest parking lot is full. Another lot is available. It will save time, but is one block farther, and you'll be carrying groceries." (Low alignment condition) or "The nearest parking lot is full. Another lot is available. It's one block farther, but will save time, and you're running late." (High alignment condition)
Initiating Action	Low (Suggests)	"Should we reroute to that lot?"
	High (Acts)	"System rerouting to that lot."

To create scenes depicting these scenarios, we video recorded a drive through San Francisco from the driver's perspective and selected two clips. The grocery clip was 32 s long, and the work clip was 27 s long. Each was prefaced in text and spoken audio with a description of the scene and the user's primary goal (shown in the top two rows of Table 3). We overlaid an interactive navigation map onto the right side of the frame, representing the route for that scene and serving as a proactive system interface (see Figure 1). The map showed the driving route, vehicle location, destination, and the original and alternative parking lots.

5.1.2. Communication of context awareness

In addition to goal alignment, we manipulated the level of contextual awareness that the vehicle communicated to the driver. In the low awareness condition, the system communicated information only about the driving environment (the nearest parking lot is full, another lot further away is available), while in the high awareness condition, it also communicated information about the driver's task and priorities, demonstrating a knowledge of the driver's needs. The high awareness condition also used different word order to emphasize the system's low versus high alignment with the driver's goals. For example, the low alignment, high awareness condition first reported that the alternative lot would save time, but concluded by noting that it was further and the driver would be carrying groceries (shown in the middle two rows of Table 3). Our goal with high awareness was to add the dimension of understanding users and their needs, with examples being mobility constraints such as walking distance, time constraints such as being late, or task constraints such as carrying groceries.

5.1.3. Initiation of action

Finally, we manipulated the level of system agency. In the low agency condition, after informing the driver about the full lot near the destination and available parking in an alternate lot, the system suggested action by asking if the driver would like to reroute to the further lot with immediately available parking. In the high agency condition, the system took action by announcing that it was rerouting to the further lot (shown in the bottom two rows of Table 3).

5.2. Dependent variables and covariates

5.2.1. Agreement and acceptance

To measure the extent to which participants agreed or disagreed with the system's behavior, we asked *How strongly do you agree or disagree with the system's suggested action? / How strongly do you agree or disagree with the system's chosen action?* With responses on a 7-point Likert-type scale from strongly disagree to strongly agree. Regarding acceptance, the Technology Acceptance Model (TAM) developed by Davis (1989) and adapted by Meschtscherjakov et al. (2009) and Bader et al. (2011), and the Autonomous Vehicle Acceptance Model (AVAM) by Hewitt et al. (2019), have been used to predict acceptance of a technology through measures such as usefulness, ease of use, social influence, and safety. However, since we focused on users' acceptance of individual decisions the system makes, rather than acceptance of the technology itself, we asked the single question *Would you accept the system's suggested action or choose a different action? / Would you let the system take that action, or override it?* This provided for a binary yes or no assessment of their preference to accept or reject the system's behavior in a particular scenario.

5.2.2. User experience

In their study of AV transitions of control, Schneider et al. (2021) test how communication transparency affects user acceptance and experience, using the AVAM and the Short Version of the User Experience Questionnaire (UEQ-S) by Schrepp et al. (2017). They find that greater explanation improves users' evaluations of the system and experience. Since we look to understand communication of contextual information on user experience, we likewise use the UEQ-S. Applying a factor analysis to verify the scale with our participant group (using f_{α} within R), we decomposed the data into two factors: hedonic and pragmatic, matching Schrepp et al.'s two factor model. Hedonic interaction qualities relate to "pleasure or fun while using the product" while pragmatic qualities relate to "tasks or goals the user aims to reach when using the product" (Schrepp et al., 2017, p. 104). We then used R's `factor.scores`, applying the Thurstone scoring method (Grice, 2001), to calculate participants' scores for each factor.

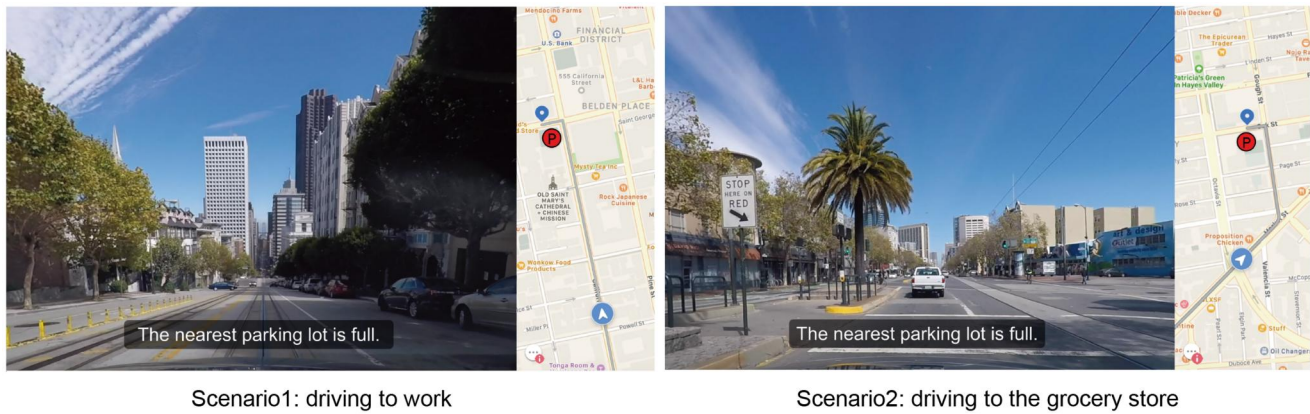


Figure 1. The two scenarios shown to participants as the study's video prompts. The left side of each panel shows the driving scene, and the right side shows the animated navigation interface.

5.2.3. Trust

Frison et al. (2019) vary AV performance and interface aesthetics and usability, and find that user experience and trust influence and correlate with each other. They suggest that research investigating trust should also “consider and include user experience measures” (p. 10), which we follow. To measure trust, we used the 12-question self-report scale developed by Jian et al. (2000). A factor analysis of the trust survey responses yielded two distinct factors, measuring trust and distrust, somewhat different than Jian’s suggested single human-machine trust factor, which had trust and distrust on opposite ends of the same scale. We then calculated participants’ factor scores for trust and distrust, using the method described above.

5.2.4. Dispositional traits

Merritt and Ilgen (2008) find that prior research pays more attention to the characteristics of automated systems than to those of the humans interacting with them. They demonstrate that individual differences affect perceptions of machine characteristics when those characteristics are constant, and emphasise the importance of human users’ individual differences when examining trust and automation use. We therefore consider how such traits predispose participants’ perceptions and covary with our other measures.

In particular, individuals with higher disposition to trust technology display greater trust in automated systems (Biros et al., 2003; Hoff & Bashir, 2015; Merritt & Ilgen, 2008), so we included the Propensity to Trust (PTT) scale developed by Frazier et al. (2013) as a covariate measure. Additionally, locus of control, defined by Özkan and Lajunen as “the degree to which a person generally perceives events to be under their own control (internal locus of control) or under the control of powerful others or other outside forces (external locus of control)” (Özkan and Lajunen, 2005, p. 534; Rotter, 1966) is likely to affect both user experience and trust in proactive systems. Since our study is in the context of vehicle interaction, we included Özkan and Lajunen’s Multidimensional Traffic Locus of Control Scale (T-LOC). For both measures, we again verified the factor structure using a factor analysis, and calculated factor scores using the method described above. For our participants, we found that

PTT consists of two factors, trust and distrust, while T-LOC comprises four factors, grouped as internal (self) or external (the vehicle and the environment, other drivers, and fate) loci. We scored participant responses on each of these factors, again using the preceding method.

5.3. Procedure

We developed the study as an online video survey, advertised to participants on Prolific and administered through Qualtrics.² The survey was structured into four sections: an introduction, an elicitation of dispositional traits, the video stimulus with follow-up questions, and a brief demographic questionnaire.

In the introductory section we first framed the study’s context, and asked participants to use a device with 12-inch or larger screen. We encouraged them to watch the video clips at full screen to maximize the feeling of real driving experience. The introduction concluded with video and audio tests to ensure that they could see, hear, and understand the study’s video cues.

In the dispositional trait section of the questionnaire, participants completed a battery of questions to determine their locus of control, adapted from Özkan and Lajunen (2005); propensity to trust, adapted from Frazier et al. (2013); and trust in automation adapted from Jian et al. (2000).

Then participants were primed to think about a recent driving and parking experience. Having participants write about their experience is a highly effective method of priming, stronger than just reading a vignette. Participants were instructed to describe a recent situation where they had to carry packages, and/or a situation where they were under time pressure. They wrote detailed and sometimes humorous answers, including statements about not wanting to get rained on, or wanting to park at the far end of the parking lot to get in a few more steps of exercise. This was followed by a set of sensitizing questions about their recent parking experiences, to place them in a mental state related to driving, navigation, and parking.

The video stimulus section opened with a text and image explanation of the scene’s navigation interface, shown in Figure 2. Participants were randomly assigned to one of the

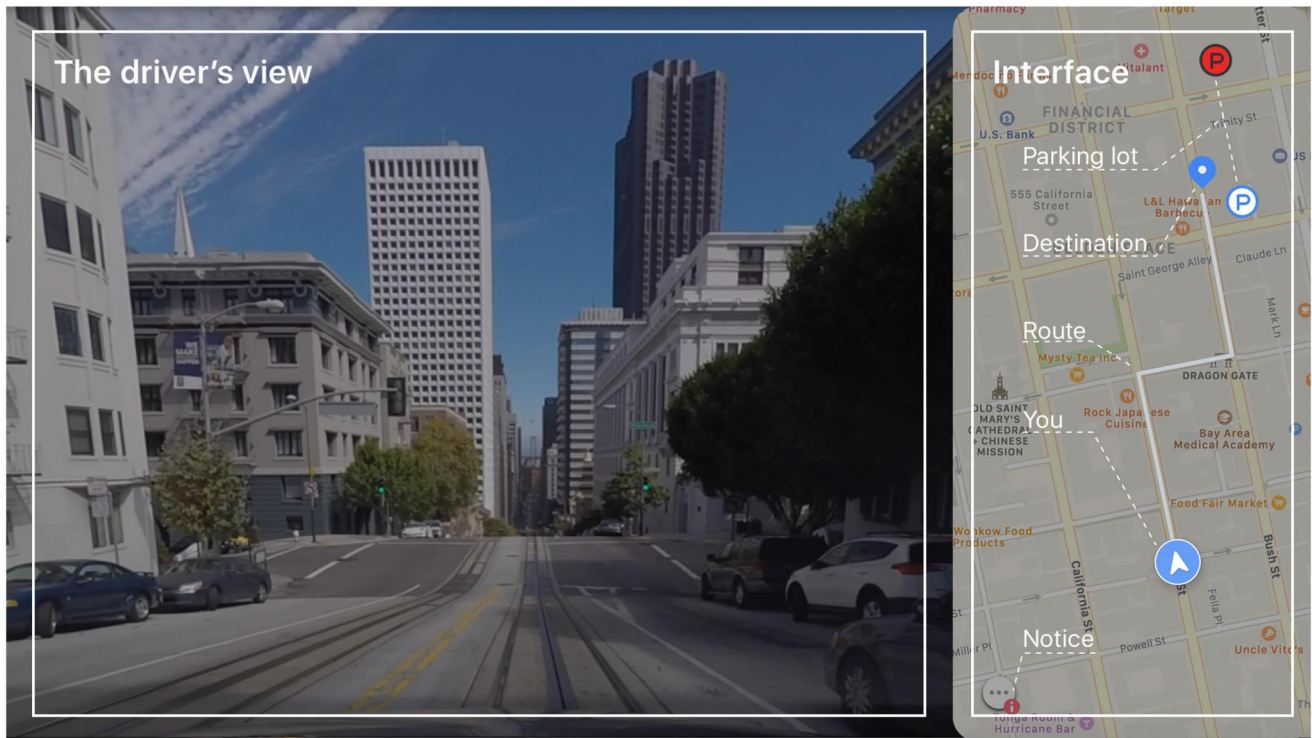


Figure 2. Instruction page, with still image of the user interface, shown to participants prior to viewing the scenario video.

eight conditions and then viewed the corresponding video. Each video opened with a description of the scene shown in text and read by a synthesized male voice. These descriptions situated participants in the scenario and established their goal of minimizing either time or distance.

Once the video of the drive began, and while the car was still a few blocks from the destination (about 15 s into the clip), an audio alert sounded. Immediately following, the navigation system informed participants (in a synthesized female voice) that the closest parking lot was full, accompanied by parking lot icon animations showing the changed state of both parking lots. In the high context awareness condition, the system also communicated the additional information about the driver's task and priorities. Finally, depending on agency level, the system either asked the driver's permission to reroute to the alternative lot, or informed the driver that it was doing so.

After watching the video, as an attention check, we asked participants to describe the action that the system suggested or took. They then answered a set of questions about how strongly they agreed or disagreed with the system's action or suggestion, whether they would accept that action or the suggested action, and why. Then they filled out questionnaires on both user experience and trust in automated systems.

To close the questionnaire, participants completed a set of demographic questions asking about their experience, frequency, and ability in driving, their experience with partially automated driving systems, and their age and gender. To check response quality and verify that participants took sufficient time to watch the video and answer questions, we tracked their time on every page throughout the survey.

Participants took 10 min on average ($M = 10.5, SD = 5.4$) to complete the study, not including one outlier whose time was 159 min, likely due to taking a break during the session.

5.4. Participants

We recruited 400 participants, with 351 remaining after filtering out those whose response times and patterns were insufficient to have watched the video stimulus or answered questions with due diligence. They were fairly evenly distributed among conditions (see Table 4), and ranged in age from 18 to 77 years ($M = 33.6, SD = 12.6$), with 157 females, 185 males, 7 others, and 2 who did not answer this question.

5.5. Hypothesis testing

In order to avoid overfitting, we conducted a limited set of statistical analyses, tuned to test our specific hypotheses. We therefore conducted a hierarchical regression analysis, building models stepwise to evaluate the effects of the multiple components. The factors include the two trait measures (propensity to trust and locus of control), and the three independent variables, representing our dimensions of proactivity (goal alignment, context awareness, and initiation of action). This subsection describes these statistical analyses.

5.5.1. H1: Direct effects of dimensions

To test if independently varying each of our three dimensions could improve agreement, acceptance, user experience, and trust scores (RQ3, H1), we analyzed the interval responses from the Likert-type scaled questions as linear

Table 4. Participant count (*N*), mean age (yrs), and driving experience (yrs) across all experiment conditions.

Goal Alignment	Context Awareness	Initiating Action	<i>N</i>	Age <i>M</i> (<i>SD</i>)	Yrs. Driving <i>M</i> (<i>SD</i>)
Low	Low	Suggest	43	35.0 (12.2)	16.9 (11.5)
Low	Low	Act	47	33.2 (10.1)	15.6 (9.7)
Low	High	Suggest	45	35.1 (14.8)	17.6 (14.1)
Low	High	Act	41	32.9 (11.4)	15.4 (11.8)
High	Low	Suggest	43	33.7 (13.6)	15.8 (13.1)
High	Low	Act	45	30.6 (11.3)	13.4 (11.2)
High	High	Suggest	45	34.9 (13.6)	17.5 (14.3)
High	High	Act	42	33.6 (13.7)	16.6 (13.9)

models (using `lm` in R) and the dichotomous responses from the acceptance question using hierarchical binomial logistic regressions (using the binomial generalized linear model `glm` in R).

5.5.2. H2: Interaction effects of dimensions

Next, to test if there are significant interactions between the dimensions (RQ3, H2), we modified the models above to include pairwise interactions among the three dimensions: (goal alignment * context awareness), (goal alignment * initiating action), and (context awareness * initiating action).

5.5.3. H3: Effects of dispositional traits

Finally, to check if the trait factors affect measures user experience and trust (RQ4, H4), we included PTT and the internal (self) and external (the car and the environment) loci from T-LOC scales as covariates in the H1 models.

6. Results

6.1. Dependent variables

Figures 3–5 show participants' scores for survey measures of agreement with and acceptance of the system's behavior, hedonic and pragmatic components of user experience, and trust and distrust. Within each figure, each panel represents one condition from our category model, having high or low alignment with user needs or goals, communication awareness of context, and agency and taking initiative.

6.2. Hypotheses outcomes

The significance test results for the three hypotheses are included in Tables 5–7.

6.2.1. H1: Direct effects of dimensions

There were significant direct effects of (a) *alignment between the system's behavior and the user's goals* on agreement with and acceptance of the system's behavior, the hedonic component of user experience, and trust, supporting H1(a) for these measures. There were also significant direct effects of (b) *communication of the system's awareness* on agreement, acceptance, and trust. There were no significant direct effects of (c) *initiation of action* on any of the dependent measures, thus providing no support for H1(c).

6.2.2. H2: Interaction effects of dimensions

There were significant interaction effects of (a) *goal alignment * context awareness* on agreement, acceptance, and the hedonic factor of user experience, thus partially supporting H2(a). No significant interaction effects were found for (b) *goal alignment * initiation of action* or (c) *context awareness * initiation of action*, providing no support for H2(b) nor H2(c).

6.2.3. H3: Direct effects of dispositional traits

For (a) *direct effects*, we found propensity to trust to be a highly significant predictor of participants' scores on trust, distrust, and both hedonic and pragmatic components of user experience. There were significant main effects for the external (vehicle and the environment) but not the internal (self) loci of T-LOC, on both hedonic and pragmatic components of user experience, and no significant effect on other measures. For (b) *interaction effects*, there were no significant interactions between PTT or T-LOC and any of the three dimensions on the dependent measures.

6.3. Qualitative responses

We coded and analyzed participants' rationales for accepting or rejecting the system's decision or suggested action, to check if their choices made sense given the goal of their scenario. To do so, we compared the textual responses for two survey questions: (1) the pre-video question asking participants to recount a recent driving and parking experience, and (2) the post-video question asking why they might or might not allow the system to take action or accept its suggested action. We followed an open coding process (Khandkar, 2009), labeling and grouping thematically similar answers, and derived 16 codes to represent participants' rationales for their choices. The codes and their frequencies are shown in Table 8.

6.3.1. Work scenario

The Work scenario aligned the system's choice with the participant's goal to minimize travel time. Participants accepted the system's action/suggestion 93% of the time, generally citing three reasons: (1) they were concerned about time (39% of responses), for example, "I don't want to spend the time trying to find another parking space when I'm already running late," (2) they trusted it (18%), specifically using the word "trust" in most cases, and (3) the

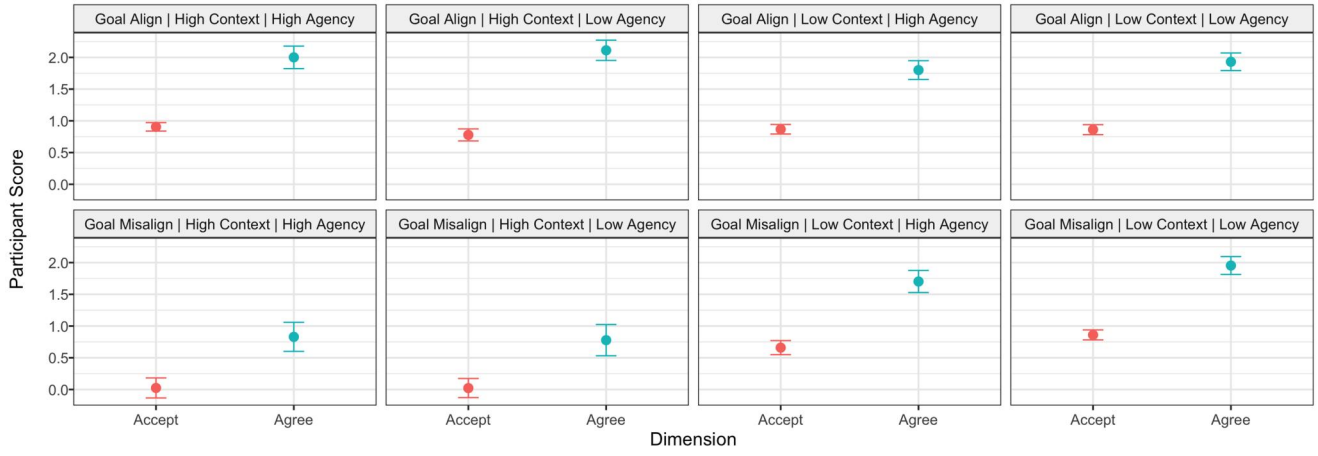


Figure 3. Participant scores for agreement with and acceptance of the system's behavior. The top row is goal aligned and the bottom is misaligned; the 2 left columns are high communication of context and the 2 right are low; alternate columns are high and low agency.

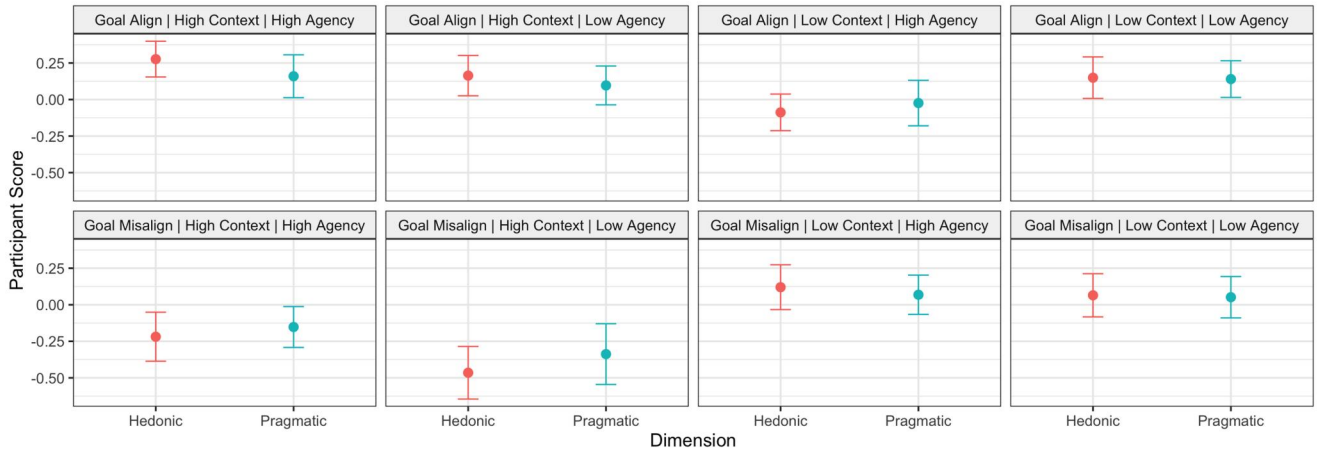


Figure 4. Participant scores for the hedonic and pragmatic components of user experience. The top row is goal aligned and the bottom is misaligned; the 2 left columns are high communication of context and the 2 right are low; alternate columns are high and low agency.

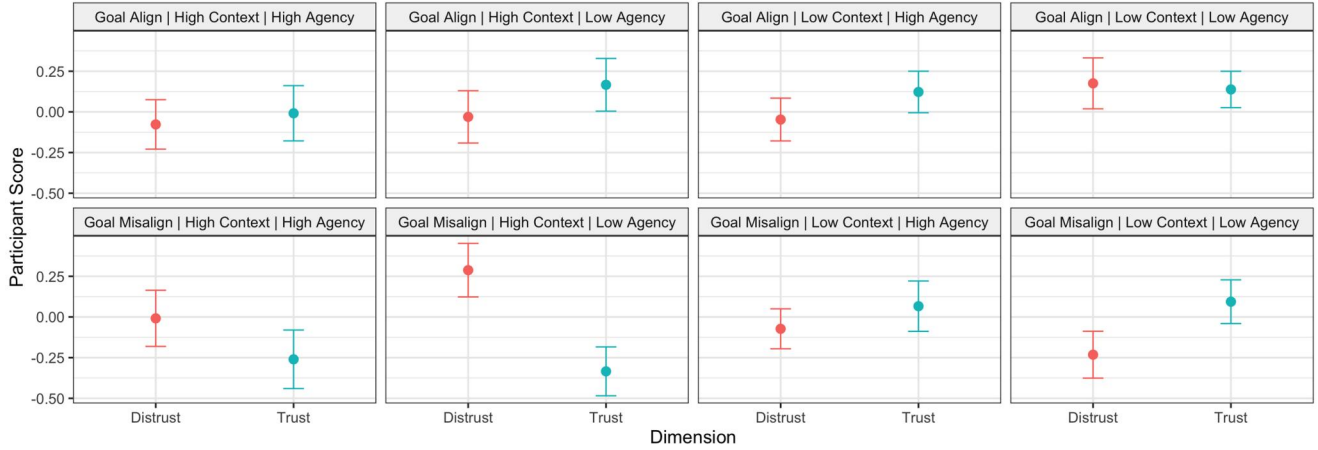


Figure 5. Participant scores for trust and distrust measures. The top row is goal aligned and the bottom is misaligned; the 2 left columns are high communication of context and the 2 right are low; alternate columns are high and low agency.

choice seemed like a good one and made sense (17%), with participants using terms such as “a better choice,” “logical,” and “made sense.”

Only 7% of participants rejected the system's choice. In these cases, rationale was varied, with the most cited being (1) they were concerned with distance they would need to travel (2%), giving reasons such as “I would first check for a

better alternative that might be closer, e.g., on street parking.” or “Because I am stubborn and would prefer a place closer,” (2) they doubted the lot was actually full (2%), saying “Because the parking lot is close to my final destination, and I just want to make sure that the parking lot is full.” or “I would probably double check to make sure the lot is actually full,” and (3) they did not fully trust the system (1%), for

Table 5. Results of hypothesis testing for H1's direct effects.

Measure	H1(a) Goal Alignment			H1(b) Context Awareness			H1(c) Initiating Action		
	<i>b</i>	<i>t</i> or <i>z</i>	<i>p</i>	<i>b</i>	<i>t</i> or <i>z</i>	<i>p</i>	<i>b</i>	<i>t</i> or <i>z</i>	<i>p</i>
Agreement	0.64	4.80	<.001	−0.41	−3.03	.003	−0.09	−0.68	.496
Acceptance	1.68	5.07	<.001	−1.32	−4.32	<.001	−0.01	−0.03	.980
Hedonic UX	0.24	0.11	.022	−0.12	−1.17	.245	0.05	0.46	.643
Pragmatic UX	0.18	1.71	.088	−0.12	−1.11	.268	0.03	0.28	.781
Trust	0.21	1.99	.047	−0.21	−1.99	.047	−0.03	−0.26	.798
Distrust	0.01	0.09	.932	0.09	0.85	.395	−0.10	−0.98	.330

Linear models show *t* values and dichotomous models show *z* values. Bold results are significant at the 5% level or greater.

Table 6. Results of hypothesis testing for H2's interaction effects.

Measure	H2(a) Alignment*Context			H2(b) Alignment*Agency			H2(c) Context*Agency		
	<i>b</i>	<i>t</i> or <i>z</i>	<i>p</i>	<i>b</i>	<i>t</i> or <i>z</i>	<i>p</i>	<i>b</i>	<i>t</i> or <i>z</i>	<i>p</i>
Agreement	1.21	4.77	<.001	−0.07	−0.26	.794	0.17	0.62	.533
Acceptance	1.75	2.51	.012	0.66	0.97	.332	0.77	1.22	.224
Hedonic UX	0.63	3.01	.003	−0.24	−1.12	.265	0.27	1.27	.205
Pragmatic UX	0.38	1.79	.074	−0.17	−0.78	.436	0.20	0.93	.354
Trust	0.33	1.56	.119	−0.13	−0.62	.534	−0.03	−0.13	.900
Distrust	−0.41	−1.92	.055	−0.06	−0.26	.797	−0.14	−0.65	.514

Linear models show *t* values and dichotomous models show *z* values. Bold results are significant at the 5% level or greater.

Table 7. Results of hypothesis testing for H3's dispositional traits.

Measure	H3 PTT			H3 T-LOC (Self)			H3 T-LOC (Environment)		
	<i>b</i>	<i>t</i> or <i>z</i>	<i>p</i>	<i>b</i>	<i>t</i> or <i>z</i>	<i>p</i>	<i>b</i>	<i>t</i> or <i>z</i>	<i>p</i>
Agreement	0.33	4.67	<.001	0.02	0.26	.793	0.09	1.27	.205
Acceptance	0.67	4.65	<.001	0.02	0.14	.889	−0.02	−0.12	.906
Hedonic UX	0.29	5.35	<.001	0.04	0.71	.481	0.15	2.78	.006
Pragmatic UX	0.24	4.36	<.001	0.05	0.98	.329	0.12	2.33	.020
Trust	0.50	10.13	<.001	0.05	0.90	.371	0.10	1.83	.068
Distrust	−0.29	−5.27	<.001	0.10	1.88	.061	−0.001	−0.02	.981

Linear models show *t* values and dichotomous models show *z* values. Bold results are significant at the 5% level or lesser.

Table 8. Codes representing participants' rationales for accepting or rejecting the system's action or suggested action and their frequencies by condition.

Code	Work	Accept	Reject	Grocery	Accept	Reject
Save time	94	93	1	41	40	1
Trust	44	43	1	39	39	0
Makes sense	41	41	0	41	39	2
Reduce risk	14	14	0	4	4	0
Distance	12	7	5	38	11	27
Limited information	7	7	0	11	11	0
Verify if lot is full	5	0	5	11	0	11
Save stress	5	5	0	9	9	0
Distrust	3	1	2	8	0	8
Don't mind carrying/waiting	3	3	0	5	5	0
Less bother	3	3	0	4	4	0
Test that system works	2	1	1	2	0	2
Prefer spot immediately	2	2	0	4	3	1
No other option	2	2	0	5	5	0
Things can change	1	1	0	12	0	12
Prefer to do it myself	1	0	1	2	0	2
	239	223	16	236	170	66

The Work and Grocery totals are a sum of the adjacent Accept and Reject counts.

example, "I would not feel comfortable following an automated system for all decisions."

6.3.2. Grocery scenario

The Grocery scenario misaligned the system's choice with the participant's goal to minimize walking distance. Participants accepted the system's action/suggestion 72% of the time, again broadly citing three reasons: (1) they were concerned about time (17% of responses), using terms such

as "to save time" and "don't want to waste time," (2) they trusted it (17%), for example, saying "I would trust what the system was talking about and go with their suggestion." or "In that certain scenario, the system knows better than me." or "If I didn't know the area, I think I would probably do whatever my car robot told me to do," and (3) the choice made sense (17%), with rationales such as "It made sense and was not that much further." or "I assume the system is integrated with the parking vacancy status and knows better than I do whether or not the parking lot is open." or "The nearest parking lot was full. So, this is a reasonable solution. I just have to suck it up and walk a little longer."

About 28% of participants rejected the system's choice. In these cases, they: (1) were concerned about carrying groceries a greater distance (11%), (2) doubted whether the lot was actually full (5%), such as "I would want to see if the parking lot was actually full." or (3) thought the situation might change by the time they arrive (5%), saying "Grocery lots have quick turnover. I'd rather wait a few minutes for a spot to open." or "Someone might be leaving so we might be able to get a space." or "Parking spaces become available or get taken in a flash, I would drive around said first lot until a spot was open."

7. Discussion

Our review of prior work provided a foundation for answering RQ1 and RQ2, informing our approach to proactivity

and choice of specific dimensions that may impact proactive interactions. These dimensions were then applied and tested in an experiment that varied dimensions of goal alignment, communication of contextual awareness, and level of agency in a proactive navigation system.

Participants' responses to the system's actions provide empirical support for the three dimensions we investigate being relevant to the design of proactive systems. These factors do not all have significant direct effects on driver behavior, but significant interaction effects are present. This indicates that the system can and should be designed to communicate and behave in a manner that elicits calibrated trust and justified reliance, effectively aligning itself with drivers' goals. Participants' qualitative responses support the behavioral findings, regarding trust in the system being influenced by manipulation of the three dimensions of our model. These findings suggest specific design characteristics that should be followed to best influence trust and reliance, based on the system's actual capabilities.

7.1. Revisiting the hypotheses

7.1.1. H1: Direct effects of goal alignment, quantity of communication, and initiation of action

(a) We expect, and the data in Table 5 show, that goal alignment is the most salient and relevant of the three dimensions to users' agreement with and acceptance of the system's actions and to their quality of experience. Drivers naturally focus on goals such as arriving to a meeting on time or minimizing the distance to carry groceries during an outing, and are thus likely to acutely feel differences when the system either supports or contradicts these goals. Their sense of working together with or against the system could then influence their perception of self-similarity (Hommel & Colzato, 2015; Huang et al., 2020) with the system and thus influence feelings of trust.

7.1.1.1. Design implication. To optimize acceptance of the system's actions, user experience, and feelings of trust, proactive systems should strive to understand their users' needs and/or goals and align their behavior toward supporting users' goals, as conditions or other priorities or constraints permit.

(b) Increasing communication decreased every measure except distrust, which is consistent with it being in principle the opposite of trust, as shown in Table 6. This is in contrast to our expectations that increased communication of the system's awareness of the environment would increase trust, thus our findings do not support H1b. Evaluating interaction effects, Figures 3 and 5 show that this outcome is driven by the goal misalignment condition. When goals are aligned (the top row of each figure), greater communication of context awareness has little influence on these measures, but when goals are misaligned (the bottom row of each figure), there is a notable negative difference. This makes sense, because in this condition, greater communication highlights the fact that the system's action or recommended

action is contrary to the driver's goals, and implies that the system is aware of this.

7.1.1.2. Design implication. Reeves and Nass (1996), note that "each speaker in an interaction should contribute only what the conversation demands, not too much or too little" (p. 29). This agrees with our finding that communicating more contextual information can lead to negative responses in some conditions, such as when goals are misaligned, but not others. Therefore, if the appropriate level of informing is uncertain, it may be preferable to bias toward communicating too little rather than too much.

On the other hand, when the system's behavior *appears* to differ from the user's goal, such as choosing a parking lot one block farther away, but *in reality* supports the goal, such as saving time (the first and third on top in Figure 3), communicating more contextual information could reduce the potential for misunderstanding between humans and systems in cooperation.

(c) Being more proactive, independently initiating action rather than responding to user behavior did not significantly affect agreement with the system or acceptability of its behavior, thus showing a lack of support for H1c in terms of the direct effect of proactivity. Possible explanations for the lack of support for H1c include: (1) that the difference between suggesting and taking action was too subtle when compared to the difference between either one and no agency, (2) that there was too little communication of context to highlight the difference, or (3) that greater agency, of itself, does not impart a positive or negative influence on our measures of social interaction. The first may change over time, as automation becomes more common and drivers become more attuned to different degrees of agency, while the last is consistent with our (earlier) premise that goal alignment and communication of context may exert greater influence than agency on perceptions.

7.1.1.3. Design implication. The lack of significant difference suggests that there may be little practical difference between suggesting and taking action (with respect to the included measures). If so, then it may be unnecessary (and a greater interruption) to request the user's approval prior to every action. Situations demanding a rapid response, such as emergency situations, may not allow for waiting for driver input, and further study will be needed to understand and establish appropriate relationships between levels of safety and initiation of action in such situations. Even in less pressured situations, however, proactive behavior may be well received provided that there is no pronounced conflict between system action and driver goals.

7.1.2. H2: Interaction effects

While goal alignment on its own may be the most salient and relevant dimension, calling to the user's attention (through increased communication) when the system either acted in alignment with or contrary to their goals intensified responses to these three measures in both positive and negative directions.

7.1.2.1. Design implication. It is not always possible to achieve goal alignment, as competing requirements such as safety may take priority in a given situation. In these cases, crafting communication may be the strongest tool at the designer's disposal to optimize user experience.

7.1.3. H3: Effects of dispositional traits

We expected an interaction between PTT and goal alignment, as both had a strong effect on trust, but regression analysis did not show a significant interaction effect. PTT and goal alignment account for most of the variance in self-reported trust ($b = 0.50$ and $b = 0.21$ respectively). While the data did show some interaction between PTT and goal alignment ($b = 0.12$), it was not a significant component of the regression, accounting for only a small part of the variance.

7.1.3.1. Design implication. As dispositional traits such as propensity to trust and locus of control impact perceived user experience and trust, personalization through adaptive or adaptable automation, such as accounting for users' preferred driving styles (Basu et al., 2017) or sensation seeking (Peng et al., 2024; Yusof et al., 2016) may be a worthwhile approach to improve the design of proactive systems.

7.2. Qualitative responses

7.2.1. Work scenario

Participants' rationales for accepting the system's choice in the Work scenario were in keeping with our expectations, with participants describing the system as aligned with their interests, trustworthy, and sensible, so they went along with its choices. The number of rejections was low, which makes sense given the system's expression of context awareness and alignment with the driver's goals. But some participants still wanted to check if they could park closer, even though they would not be carrying bundles like in the Grocery scenario. Perhaps given their (real world) experience with the rate of parking turnover, participants retained some skepticism that the system could be up-to-date or correct about current spot availability.

7.2.2. Grocery scenario

Participants' top three rationales for accepting the system's choice in the Grocery scenario were the same as those in the Work scenario. This surprised us, particularly since a large portion of participants expressed them, and the system's choice contradicted their goal in this scenario. We set up the scenario to emphasize walking distance, but participants seem to have other preferences or interests that were unrelated to the goals of the scenario and that were of greater priority to them. Other than being most concerned about carrying groceries, participants' rationales for rejecting the system's choice were similar to those in the Work scenario, and their quotes support our idea above that drivers have varied interests, as well as a lifetime of experience parking at grocery stores, which they likely trust and can

draw inferences from more than a report of current parking conditions.

7.2.3. Additional insights

Regardless of the scenario, participants' decisions to accept the system's behavior were driven largely by the same three factors: to save time, their trust in the system, and whether the action made sense to them. This provides insight into drivers' primary concerns about parking in different scenarios, and suggests that if a proactive system wants to motivate acceptance (for example, when it cannot meet the stated goal) it might need to focus on factors such as these, regardless of the type of scenario.

On the other hand, participants' primary concerns when rejecting the system's behavior were different than the top three concerns when accepting its behavior. This implies a non-symmetrical nature to drivers' decision-making, where prioritization of different factors led to a rejection instead of an acceptance, rather than just whether a particular factor is or is not satisfied.

The emphasis participants put on the action making sense suggests that system transparency, in providing an explanation, might help influence users to accept vehicle behavior whether or not it is aligned with the goal of the scenario and the ability to understand the reasoning behind AV suggestions. Furthermore, focusing the explanation on the likelihood to save time, when applicable, could motivate acceptance when other goals, such as minimizing distance, cannot be met.

The breadth and sometimes unexpected nature of other concerns in participants' rationales highlight the complexity of user decision-making and the influence of factors that may contradict or be unrelated to scenario-specific goals and constraints. This is further evidenced where some preferred to rely on their own knowledge and experience rather than cede decision-making authority to the vehicle in all situations.

8. Limitations and future work

The situated study to explore our proactive framework is subject to some limitations, which can also be viewed as opportunities for further work.

To establish context, the study used a single video stimulus. While this format does not afford the same ecological validity as a high presence simulation or on-road study, it does offer access to a large and diverse participant sample. A follow-up study in either of these environments would complement the current work. Also, by exploring a single interaction, findings are limited to participants' initial experiences and trust evaluations, rather than those developed over repeated interactions, which would be the likely use scenario. A study that ran for a longer time, or involved multiple sessions, would likely produce findings closer to users' real-world, evolving responses.

Regarding design, the study employed a relatively mild manipulation of agency: wording that either suggested or

reported a change in course. A stronger manipulation, such as following through and implementing the course change in the video, may better distinguish conditions. Also, subjective measures of trust or reliance do not necessarily evaluate participants' *actual* trust or reliance, rather their assessment of them. Studying reliance directly would require behavioral measures of user' reactions, for example through braking or steering, to actions misaligned with their goals. Finally, our system implied certainty in its communication of context awareness, whereas real systems are subject to uncertainties in sensing and analyzing conditions. Exploring whether or how to represent such uncertainty has seen little research, especially in the context of automated vehicles, and would be a natural follow-up to the current study.

9. Conclusion

We derived a framework comprising (1) initiation of action, (2) anticipation of users' needs and goals, and (3) communication of situation awareness, which draws from and aligns with prior research on proactivity across disciplines, to guide researchers and designers studying social interaction with automated vehicles. A study demonstrating the framework in use in the context of vehicle navigation led to insights that include:

- To support favorable user experience and properly calibrated trust, proactive systems should understand and align their behavior with their users' needs and goals.
- Communicating awareness of the situation ahead can reinforce both positive and negative effects of acting on aligned or misaligned goals. But since it is not always possible to align behavior with users' (immediate) goals, systems should communicate strategically.
- If suggesting versus initiating action does not lead to different user responses, it may be best to choose the less intrusive option of taking action, rather than asking users each time the system makes a decision.
- Users' dispositional trust and locus of control (to a lesser extent) affect their agreement with and acceptance of a system's choices, experience, and trust, so it is important to account for differences in traits when designing adaptive behaviors.
- Due to the breadth of unique user interests and preferences in various situations, rule-driven decision systems should be supplemented by opportunities for users to modify or personalize system behavior, or for systems to learn these preferences.

Notes

1. P to the capacity to sense states and effect aspects of the environment (Kaber, 2018), to making decisions and choices (Nyholm, 2018), as a few examples.
2. The Prolific participant recruitment platform is at <https://www.prolific.com>, and the Qualtrics online survey platform is at <https://www.qualtrics.com>.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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